Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - II
**Course Code : ** CSE 0533 1201 (CSE) & PHY 0533 1201 (EEE)

Term: Spring – 2024	**Date: 22.01.2024**	**Time: 03.30 PM**	**Room: 327**	**Activity: Lecture 7**

Topics to be discussed: Reversible & irreversible processes, Second law of thermodynamics, Carnot; Efficiency of heat engines, Carnot theorem

Target CLO & PLO : CLO1,23 & PLO1,5,9

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define Reversible and Irreversible processes
- ❖ State & Explain Second Law of Thermodynamics
- ❖ Define Efficiency of Heat Engine
- ❖ Draw & explain the Carnot's cycle.
- ❖ Solve assigned recommended mathematical problems

Methods of Teaching:

Mode/Activities	**Time**
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

1) What is the unit of efficiency?
   a) N b) m c) F d) Nm^{-1} e) none
2) What is correct condition between source (T_1) and sink (T_2) temperature?
   a) $T_1 < T_2$ b) $T_1 > T_2$ c) $T_1 = T_2$ d) $T_1 \leq T_2$ e) $T_1 = T_2 = 0$

Sample Questions (SAQ):

1. State the Second Law of Thermodynamics.
2. What do you mean by Reversible and Irreversible processes? Write with Examples?
3. What is Heat Engine? Draw a block diagram of Heat Engine.
4. Define Efficiency of Heat Engine. Find the equation of efficiency of Heat Engine.
5. Is it possible to get an engine with 100% efficiency? -Explain.
6. What are the ideas you can make a good heat engine from efficiency of engine?
7. Draw & explain the Carnot's cycle.
8. A heat engine takes 840J heat energy from the source and rejects heat energy of 630J to the cold sink. Find the efficiency of the heat engine.
9. A heat engine operated between Source with 600K temperature and sink with 340K temperature. It absorbs 950J heat energy from source. Calculate the efficiency and work done for one cycle of the engine.

Reference Books: 1. Giasuddin ahmed – Physics for Engineers (Part-1)
2. Halliday Resnick - Fundamental of Physics, 2nd edition

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 01-64
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 01-36